

Musa SAHIN

SENIOR SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Software Engineer with 9 years of experience. Primary focus on **Flutter (7 years)** and **NestJS (2 years)**, with additional practical experience in native **iOS (Swift)**, **Android (Kotlin)**, and **React Native**. Experienced in building scalable architectures, implementing mobile security standards, and establishing automated testing workflows through **TDD** & Clean Architecture. Specialized in **FinTech**, **Geomatics**, **E-commerce**, **Education**, and **Gaming** domains. Conducted thesis research on **NLP** (Natural Language Processing). Implements algorithms directly from research papers (as seen in **Guardrail**) to build custom AI agents and autonomous systems.

WORK EXPERIENCE

Team Lead & Senior Developer Orion Innovation (Edison, NJ - USA Remote)	Oct 2024 - Present
DeerCast (Mobile & Web) — Team Lead (Jan 2026 - Present): - Leading the cross-functional development team for DeerCast, a premier hunting prediction and media platform, across Mobile (Flutter) and Web. - Contributed to the development of the AI Review Agent repository to automate code reviews and quality checks. - Responsible for end-to-end development processes, including CI/CD pipeline maintenance and application delivery. - Accelerated development workflows and efficiency by at least 4x using Agentic AI integrations. HuntStand (Mobile & Web) — Senior Developer (Oct 2024 - Jan 2026): - Developing HuntStand, a cross-platform application for Hunting and Land Management using Flutter. - Utilizing Generative AI tools to optimize code quality, automate refactoring, and accelerate feature implementation. - Updated the internal MapSDK to improve geospatial rendering performance across multiple products. - Implemented Test-Driven Development (TDD) workflows, addressing technical debt in legacy modules. - Conducting code reviews and mentoring team members on modular architecture and state management.	
Software Developer Engineerica Inc (Oviedo, FL - USA Remote)	Jul 2023 - Oct 2025
- Developed and maintained AccuBus, a mobile application for monitoring state school bus logistics. - Integrated AI-assisted workflows in the development process to enhance problem-solving speed and efficient code documentation. - Rewrote the iAccu attendance mobile and kiosk application in Flutter for cross-platform support. - Contributed to the Accudemia academic center management system using native Swift and Kotlin. - Developed AccuKiosk, a lightweight version of the attendance tracking system. - Built AccuClass, a class management and scheduling application. - Implemented My Conference Planner for event scheduling and resource management.	
Software Developer & Product Manager Optimus Bilişim (Istanbul, Turkiye)	Jan 2024 - Aug 2024
- Architected white-label stock trading applications and SDKs used by multiple financial institutions (e.g., Meksa Mobile). - Refactored legacy codebases to improve architectural modularity and maintainability. - Implemented security measures to meet penetration testing requirements for financial software. - Managed the complete application rewrite to resolve performance issues.	
Software Developer & Team Lead Innovance / Oyak Yatırım (Istanbul, Turkiye)	Mar 2023 - Jan 2024
- Led the mobile development team for the Oyak Finance Trader application. - Established architectural standards and managed the release pipeline for the Flutter codebase.	
Full Stack Software Developer Nakit Yok (Ukraine Antalya, Turkiye)	Mar 2023 - Mar 2024
- Developed NOSH, a food ordering and payment mobile application using Flutter. - Built the backend infrastructure using NestJS and PostgreSQL.	
Software Developer Aymar Tech (Istanbul, Turkiye)	Jul 2022 - Mar 2023
- Led the implementation of "Untangible", an NFT marketplace and Web3-integrated game platform using React Native and Flutter. - Integrated Web3 wallets and smart contract interactions within both React Native and Flutter environments.	
Flutter Engineer United Software / Wio Bank (Ukraine UAE Remote)	Apr 2022 - Feb 2023
- Contributed to developing both the Web and Mobile versions of Wio Business, a digital banking application. - Collaborated with distributed teams to ensure banking compliance and security.	
Principal Mobile Specialist Etiya Information Technologies (Istanbul, Turkiye)	Feb 2022 - Apr 2022
- Led development of the Trumore Super App (Official App for Togg Car) using Flutter. - Implemented features to meet automotive industry performance standards.	
Software Developer & Team Lead DOP (Istanbul, Turkiye)	Oct 2021 - Feb 2022
- Led mobile development team for Siparişim and Restoran İşim Mobil applications. - Implemented micro-frontend architecture to improve scalability across mobile platforms. - Mentored junior developers on Flutter best practices and architectural patterns. - Contributed to backend development using Python (FastAPI) and NestJS.	
Software Developer Internative Software (Istanbul, Turkiye)	Mar 2021 - Oct 2021
- Developed All4Baby, a parents and baby assistant application using Flutter. - Built Sodio, a social radio application using custom Flutter UI components. - Implemented interactive mobile UI designs focused on user engagement.	
Software Developer S'aide Financial Technologies (Istanbul, Turkiye)	Apr 2020 - Feb 2021
- Developed S'aide, a student assistant and payment application using Flutter. - Integrated payment gateway APIs for secure transaction processing. - Implemented security measures to pass penetration testing requirements for financial applications. - Built the initial backend infrastructure using NestJS and MongoDB.	
Full Stack Software Developer Freelance	Jun 2017 - Apr 2020
- Developed web and mobile applications for clients using Ionic, React, React Native, and Flutter. - Delivered custom solutions for small to mid-sized projects across various industries.	

PERSONAL PROJECTS

- Created a lightweight **2.5D game engine** for Flutter (written in **C++ & Dart**) featuring declarative widgets, physics simulation, and spatial audio.
- Successfully used to develop **10+ high-quality demo games** across mobile platforms.
- Designed a custom entity-component system (ECS) optimized for Dart and Flutter's rendering pipeline.

Guardrail

| AI Code Intelligence & Security Platform

- **Platform Overview & Impact:** Developed an advanced AI Code Intelligence platform to automate security and code quality checks. **Successfully pilot tested at Orion Innovation**, demonstrating capabilities in reducing technical debt and enhancing review workflows.
- **Agent Development & Orchestration:** Built a custom, recursive RLM framework ([arXiv:2512.24601](#)) with autonomous planning (map-reduce tasks) and self-corrections via verification loops. Designed a Multi-Agent System (Explorer, Analyst, Critic) that executes code safely within a Python REPL Sandbox.
- **RAG & Knowledge Systems:** Architected a Hybrid RAG system combining Semantic Search (ChromaDB) with Knowledge Graphs (KuzuDB). Mapped code structures (AST, function calls) to answer complex "Blast Radius" queries with 99% accuracy.
- **LLMOps & Local Deployment:** Engineered a privacy-first Local LLM Pipeline using Ollama (SLM models). Implemented Cost Optimization strategies with model routing and L1/L2 caching, ensuring efficient enterprise deployment.
- **Dataset Generation & Fine-Tuning:** Implemented a **Human-in-the-Loop** mechanism that captures developer feedback and stores it in **Graph & Vector DBs** to build high-quality datasets for future **LoRA Fine-Tuning**.
- **Enterprise Integration:** Designed agents capable of interacting with corporate ecosystems (Jira, GitHub/GitLab APIs) for automated issue tracking and CI/CD analysis, effectively bridging the gap between AI reasoning and DevOps workflows.

TECHNICAL SKILLS

MOBILE DEVELOPMENT

Flutter (7 years) Dart iOS (Swift) Android (Kotlin) React Native BLoC / Cubit

BACKEND & API

NestJS NodeJS Python (FastAPI) TypeScript RestAPI GraphQL gRPC

WebSockets RabbitMQ Redis

DATABASE

Graph Databases (KuzuDB, Neo4j) Vector Search (ChromaDB) PostgreSQL MongoDB

Firebase

CLOUD & DEVOPS

Docker CI/CD (Gitlab, Github Actions) LLMOps Pipelines Model Serving (vLLM)

AI & LLM ENGINEERING

Recursive RLM (Paper-Based) Multi-Agent Systems Hybrid RAG (Graph + Vector) Local SLMs

AI Safety & Sandbox Cost & Model Routing

METHODOLOGIES & CONCEPTS

TDD Clean Architecture Microservices SOLID Design Patterns Agile / Scrum OOP

EDUCATION

BS, Information Systems and Technologies Yeditepe University - Istanbul, Turkiye	2021
BS, Management Information Systems Anadolu University - Eskisehir, Turkiye	2020
BS, Physics (Double Major) Yeditepe University - Istanbul, Turkiye	2021
MEd, Science Teacher Education Marmara University - Istanbul, Turkiye	2020
Thesis: Utilizing NLP (Natural Language Processing) for Assessing Open-Ended Examinations	
B.E. in Chemistry Engineering Marmara University - Istanbul, Turkiye	2015
BSc, Chemistry Marmara University - Istanbul, Turkiye	

LANGUAGES

Turkish: Native English: Professional Proficiency

